TRADE INFRASTRUCTURE LTD
Sales Cowboy | Enterprise Digital & Payment Solutions

COMMERCIAL PROPOSAL

Ref: SC-PROP-2026-COOP
Date: November 8, 2026
Validity: 30 Days

PRODUCT ARCHITECTURE & COMMERCIAL PROPOSAL

Cooperative Society Contributions & Fund Pooling Platform

Cross-Platform Member Mobile App (Android & iOS via React Native / Expo) + Master Executive Admin Console (Next.js)

1. EXECUTIVE SUMMARY & OBJECTIVE

The objective of this initiative is to build a modern, automated cooperative society management, member contribution, and fund pooling platform. The solution pairs a high-performance cross-platform mobile application for members (built with **React Native / Expo** for Android and iOS) with an executive administrative web management console (built with **Next.js**).

The platform simplifies and automates periodic thrift contributions (Ajo, Esusu, and Rotating Savings), targeted project pooling (such as land acquisition, welfare funds, annual dividend accumulation, and emergency reserves), and multi-signatory fund disbursements. By digitizing inflows and linking them directly with double-entry ledgers, the solution eliminates manual bookkeeping errors, prevents contribution disputes, issues automated payment reminders, and provides complete financial transparency for cooperative executives and members alike.

2. FUNCTIONAL SYSTEM SCOPE

Module 1: Member Mobile App (React Native / Expo)

- **Authentication & KYC:** Instant mobile login via SMS OTP (Termii). Member profile onboarding with KYC documentation, photo upload, and next-of-kin records.
- **Periodic Thrift Savings:** Automated tracking of recurring daily, weekly, or monthly savings cycles with real-time balance accumulation.
- **Target Fund Pools:** Dedicated project pooling (Land Purchase, Welfare, Annual Dividend, Emergency Reserve) with personal and collective progress meters.
- **Payment Gateways:** In-app debit card and dedicated virtual bank transfer processing via Paystack and Monnify with real-time wallet crediting upon webhook confirmation.
- **Offline Deposit Desk:** Upload physical bank teller slips for members making in-person branch deposits, awaiting admin verification.
- **Digital Member Ledger:** Complete transaction history with date/category filters and single-tap PDF statement exports.
- **Push Notifications & Alerts:** Automated contribution due reminders, payment success confirmations, and broadcast executive announcements.

Module 2: Master Executive Admin Console (Next.js)

- **Membership Directory & Compliance:** Comprehensive member registry, KYC document verification queue, and one-click account activation or suspension.
- **Contribution Schemes & Pool Engine:** Create and configure target pools, periodic intervals, savings targets, and default penalty rules.
- **Automated Audit & Reconciliation:** Real-time reconciliation matching payment gateway reference IDs against member balances, eliminating manual ledger tracking.
- **Offline Deposit Verification:** Review and verify uploaded bank deposit slips and credit member balances with full audit trails.
- **Multi-Signatory Disbursement:** Configurable authorization workflows for member withdrawal requests and cooperative project payouts.
- **Executive Financial Reports:** Exportable accounting ledgers in Excel (XLSX) and CSV formats for general meetings, regulatory audits, and trustees.

3. ENTERPRISE TECHNICAL ARCHITECTURE

LAYER / COMPONENT	TECHNOLOGY SELECTED	CORE PURPOSE & ARCHITECTURE JUSTIFICATION
Member Mobile App	React Native (Expo) - TypeScript	Single cross-platform codebase compiled into native Android (APK/AAB) and iOS (IPA) binaries, providing smooth native UI performance and fast update cycles.
Admin Management Console	Next.js 14, React, Tailwind CSS	Modern, responsive web application for desktop and tablet with secure server-side authentication, real-time data tables, and rapid financial filtering.
Backend API Engine	Node.js (NestJS / Express)	Modular, robust RESTful API built on double-entry accounting principles to guarantee transaction balance integrity and prevent race conditions.
Database Layer	PostgreSQL with ACID Compliance	Relational database with strict foreign keys, transactional boundaries, and automated daily cloud snapshots for enterprise auditability.
Payment Infrastructure	Paystack & Monnify Gateway APIs	Secure tokenized card processing and automated dedicated virtual accounts with HMAC webhook signature verification.
Messaging & Alerts	Termii SMS API & Expo Push Notifications	High-deliverability SMS gateway for OTP logins and critical financial notifications, complemented by free push notifications.

4. THIRD-PARTY INFRASTRUCTURE & OPERATIONAL COSTS

The cooperative maintains direct billing and complete ownership of third-party infrastructure and service accounts:

INFRASTRUCTURE / SERVICE	RECOMMENDED PROVIDER	ESTIMATED COST	BILLING SCHEDULE
Cloud API & Managed Database Server	Render / Railway / DigitalOcean	~$5 - $10 / month	Monthly (scaled by usage)
Custom Domain & Managed SSL	Namecheap / Cloudflare	~$10 / year	Annual (Free SSL via Cloudflare)
Payment Gateway Inflows	Paystack / Monnify	1.5% (capped at ₦2,000/inflow)	Per Transaction (Zero setup or monthly fees)
SMS OTP & Alert Delivery	Termii	~₦5.00 per SMS	Per Transaction / Pre-funded credit wallet
Google Play Developer Account	Google Play Console	$25 one-time fee	Lifetime registration fee
Apple Developer Program Account	Apple Inc.	$99 / year	Annual developer membership

5. IMPLEMENTATION ROADMAP & MILESTONES (4 WEEKS)

PHASE	SCOPE OF EXECUTION	DELIVERABLE HANDOVER
Week 1	**UX/UI Design, Architecture & KYC Engine** Screen wireframing and design prototype. PostgreSQL double-entry database schema modeling. Termii SMS OTP authentication engine and member onboarding flow.	Mobile UI Figma/mockups, staging backend environment, and working OTP authentication service.
Week 2	**Contribution Pools & Payment Inflow Pipelines** Implementation of periodic thrift savings logic, target project pooling, Paystack/Monnify payment integrations, webhook listener, and offline deposit upload workflow.	Functional payment inflow engine with end-to-end sandbox card and transfer testing.
Week 3	**React Native Mobile Client Deployment** Cross-platform mobile application development for Android and iOS. Real-time member balances, automated ledger views, PDF statement generation, and push notification triggers.	Internal test builds (Android APK and iOS TestFlight/Expo staging build).
Week 4	**Executive Admin Console, Reconciliation & Go-Live** Executive management web console, offline deposit audit desk, multi-signatory disbursement workflow, comprehensive UAT, production server deployment, and app store submission.	Live cloud production deployment, finalized Android/iOS production binaries, and admin training.

6. COMMERCIAL INVESTMENT & PAYMENT SCHEDULE

Total Fixed Development Investment: ₦850,000 (Eight Hundred and Fifty Thousand Naira) covering complete development of the cross-platform React Native mobile application and the Master Executive Admin Web Console.

MILESTONE PHASE	MILESTONE DELIVERABLE CRITERIA	FEE BREAKDOWN
1. Mobilization Deposit	Technical architecture, database design, backend setup, SMS OTP authentication, and complete UI/UX screen designs for mobile and admin.	**50% (₦425,000)**
2. Functional Staging Milestone	Demonstration of working React Native mobile app with active contribution pools, live payment checkout integration, and staging admin console.	**30% (₦255,000)**
3. Production Go-Live & Store Release	Cloud deployment, end-to-end payment audit verification, Android/iOS production binaries compilation, store release submission, and executive training.	**20% (₦170,000)**

7. POST-LAUNCH WARRANTY & ACCEPTANCE

Upon production deployment and store submission, Sales Cowboy provides **thirty (30) days of complimentary technical warranty support** covering bug remediation, security surveillance, calculation validation, and executive staff onboarding.

FOR: SALES COWBOY (DEVELOPER)

Lead Systems Architect
Sales Cowboy / Trade Infrastructure LTD

FOR: COOPERATIVE SOCIETY EXECUTIVE

Authorized Executive Representative
Cooperative Society Management Committee